

Urban FoodShare: A Requirement Driven Study of a Gamified Surplus Food Redistribution Platform

**Software Requirement Analysis
SIT 773 – Task 6.4: Research Project**

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Abstract

This research investigates the issue of household level food waste and proposes Urban FoodShare, a requirement driven, gamified surplus food redistribution platform to address behavioural, technological, and logistical barriers to efficient food sharing. While existing applications focus on business or commercial surplus, Urban FoodShare targets domestic contributors by integrating expiry tracking, intelligent meal planning, and NGO scheduling support within a mobile first interface. The project applies a hybrid of traditional requirement analysis techniques and data driven enhancements, including behavioural analytics and real time feedback loops, to capture specific stakeholder needs. A comprehensive set of 11 functional, quality, and business requirements is derived through risk classification and literature backed justification. Key system modelling artefacts including use case and activity diagrams, wireframes, and validation logic illustrate the platform's user centric operation. This research aligns with Sustainable Development Goal 12.3 and offers a scalable solution to reduce edible food waste, improve NGO coordination, and foster community participation in urban food sustainability.

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1. Case Study

Household food waste constitutes a significant portion of global food loss. To address these household level inefficiencies, Urban FoodShare is conceptualized as a mobile first platform that combines AI powered expiry tracking, gamified engagement, and seamless NGO and council coordination to reduce food waste directly at the source urban homes (Santos, Sevivas, & Carvalho, 2025). Unlike existing platforms such as OLIO and Too Good To Go that centre on commercial surplus redistribution, Urban FoodShare is specifically designed to resolve household behavioural and logistical barriers.

Studies show that consumer level food waste accounts for nearly 60% of all avoidable food loss, yet most digital interventions lack automation, personalization, or behavioural nudges. Recent advancements in Internet of Things (IoT) and cloud computing have demonstrated efficacy in enabling real time inventory tracking and expiry notifications, both of which are core features of the platform (Tiwari, Garg, Singh, Mishra, & Guleria, 2024). Moreover, personalized recipe suggestion systems, when combined with contextual food data, have been shown to influence usage behaviour and reduce wastage rates by over 25% in trial environments (Rostami, 2024).

Gamification further enhances user motivation and system retention. A systematic literature review by Mabalay (2025) found that gamified systems in sustainability contexts lead to sustained behaviour change, especially when integrated with measurable impact feedback and community ranking. Urban FoodShare leverages this by incorporating badge rewards, usage milestones, and progress dashboards, fostering both accountability and social participation. Additionally, research into smart food systems highlights the need for interoperability with municipal services to streamline surplus collection logistics (Wu & Lin, 2023). By offering API based integration with NGO calendars and council programs, Urban FoodShare facilitates donation scheduling without requiring manual follow ups.

The system is aligned with broader policy imperatives such as the UN Sustainable Development Goal (SDG) 12.3, which aims to halve per capita food waste by 2030 (UNEP, 2024). The Champions 12.3 2024 Progress Report identifies household food waste reduction as the most under addressed yet crucial area requiring innovation (Champions 12.3, 2024). Urban FoodShare, through its data driven, user centric design, contributes directly to this agenda by providing a scalable, tech enabled platform that empowers households to monitor, optimize, and redistribute food with minimal friction.

2. Problem and Vision Statement

2.1 Problem Statement

Urban households lack intelligent tools to manage, track, and redistribute surplus food. This results in daily waste of edible goods while nearby communities experience food insecurity. Existing platforms focus primarily on business level redistribution and neglect household level behaviour insights and expiry-based donation logistics.

2.1.1 Pain Points:

- **Lack of Household Centric Solutions:** Most existing food redistribution platforms (e.g., Too Good To Go, OLIO) focus on surplus from businesses or restaurants. There is a gap in tools tailored for household surplus management, where food waste is often more frequent but untracked.
- **No Real Time Expiry Monitoring:** Users often forget or mismanage food expiration dates due to a lack of intelligent expiry tracking. This leads to frequent disposal of edible food that could otherwise be consumed or donated.
- **Manual and Inconvenient Donation Logistics:** Households face logistical hurdles when trying to donate food such as lack of transportation, awareness of nearby NGOs, or scheduling conflicts for drop offs discouraging regular food donations.
- **Limited User Engagement in Existing Apps:** Current apps lack mechanisms to retain user interest or make food sharing a habit. Without incentives or feedback (e.g., rewards, progress tracking), user drop off rates remain high.
- **No Integration with Local Councils or NGOs:** There is limited or no support in current platforms for real time integration with municipal food programs or charity pickup schedules, making coordinated redistribution inefficient.
- **Inadequate Use of Data for Behaviour Change:** Existing platforms do not utilize behavioural data to offer personalized meal planning, food usage suggestions, or waste reduction analytics, missing an opportunity to influence user habits.

To further understand the underlying causes of the pain points associated with household level food waste and redistribution challenges, a domain analysis was conducted using a fishbone diagram.

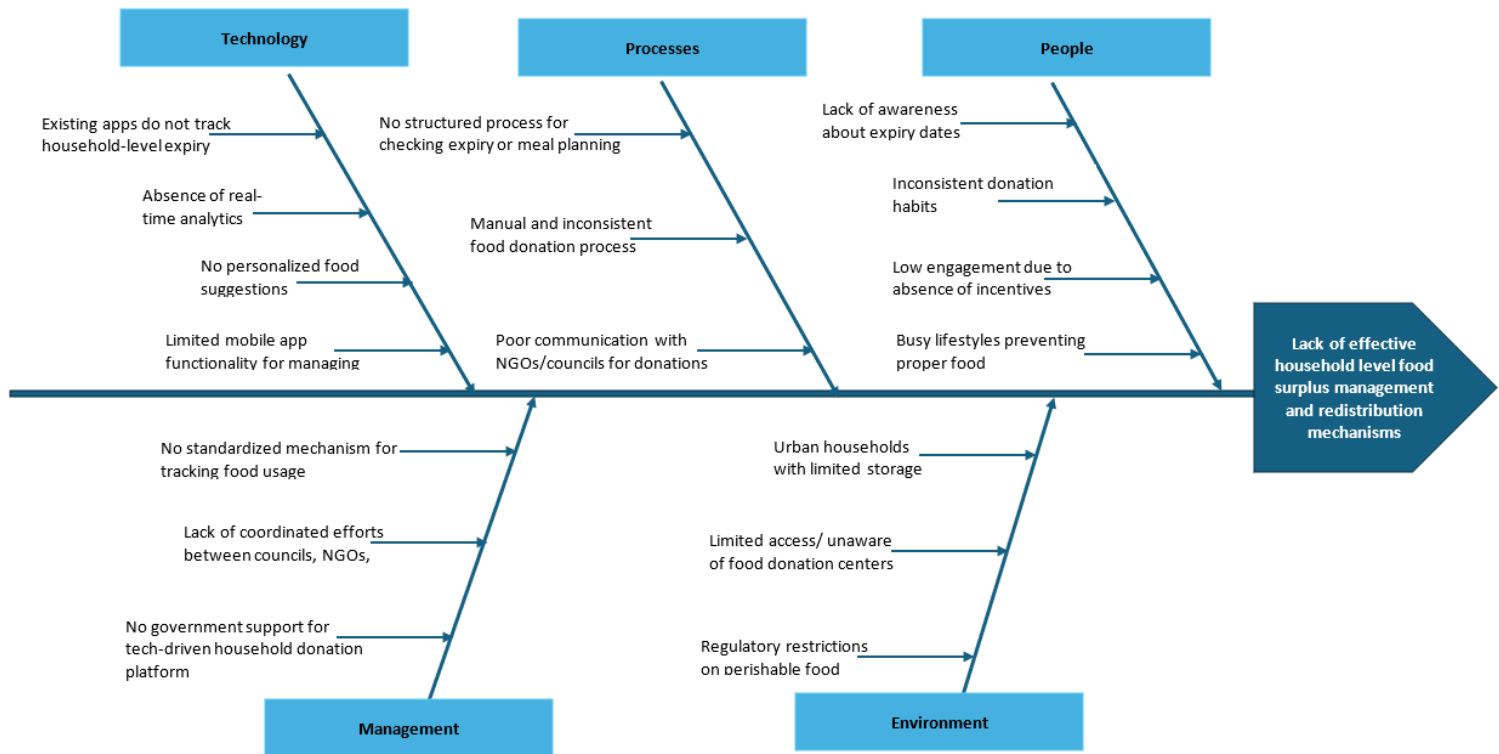


Figure 1: Fishbone Diagram

2.2 Vision Statement

Urban FoodShare envisions a mobile first, community powered platform enhanced with business analytics to help households reduce food waste by providing expiry tracking, intelligent meal planning, gamified waste tracking, and seamless integration with local NGOs and councils for scheduled pickups.

The proposed system, Urban FoodShare Platform, involves multiple stakeholders including household users, NGOs, analytics subsystems, and a recipe recommendation engine. To understand the boundary and interaction between these external entities and the system, a context diagram is presented below.

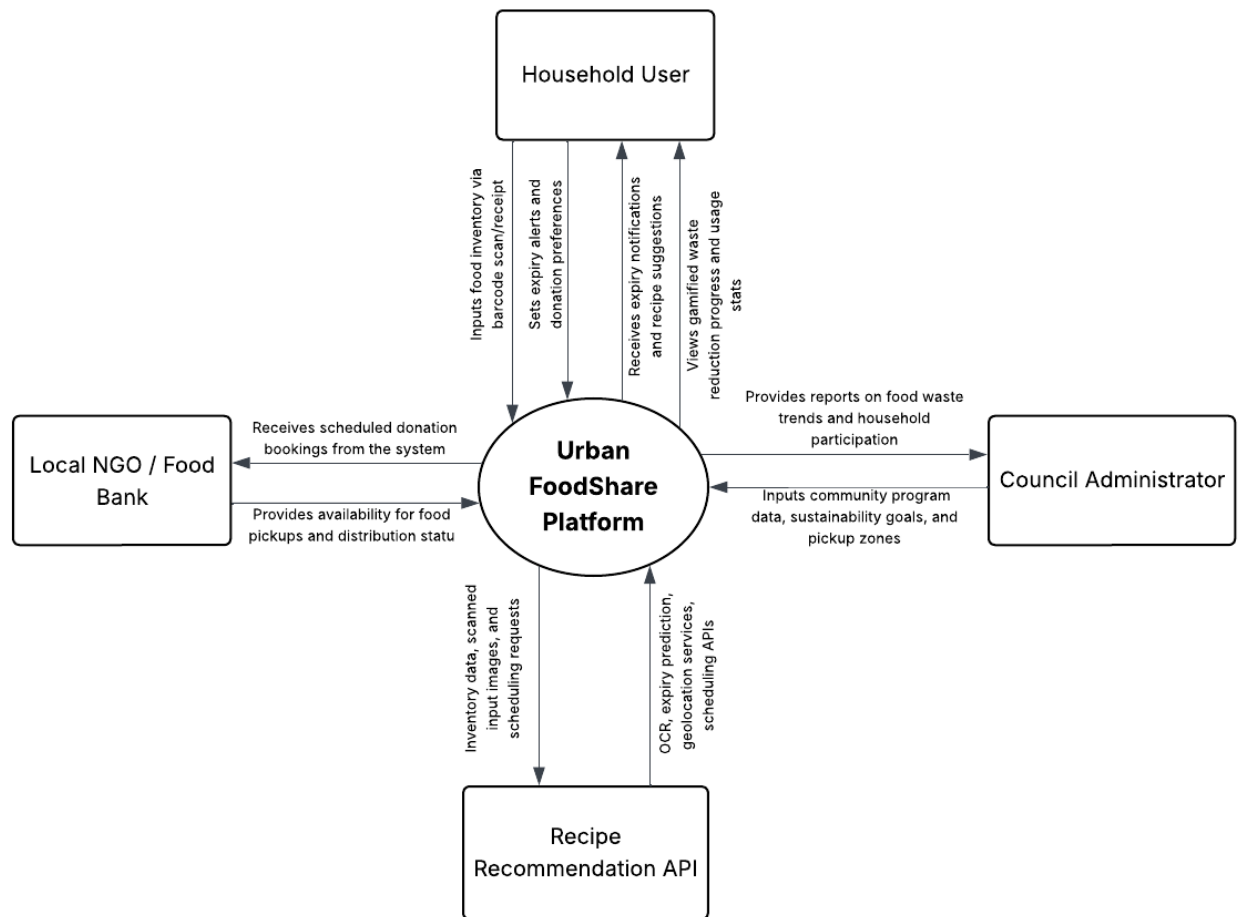


Figure 2: Context Diagram

3. Challenges in Standard Requirement Analysis Techniques

While conventional requirement analysis techniques such as interviews, user stories, and use cases are foundational for understanding baseline system functionality, they often fall short in capturing the full scope of user behaviours, motivations, and advanced analytics driven needs typical of socially conscious digital platforms like Urban FoodShare.

Table 1: Comparison of Similar App Features

Feature	OLIO	Too Good To Go	Karma	Flash-food
User-to-user sharing	Yes	No	No	No
Real-time inventory updates	Limited	Yes	Yes	Yes
Pickup scheduling	No	Yes	Yes	Yes
Expiry-based meal planning	No	No	No	No
Gamified waste tracking	No	No	No	No
NGO/Council integration	No	No	No	No

Source: (OLIO, 2023; Too Good To Go, 2025; Karma, 2025; Flashfood, 2025)

One primary limitation of these standard methods is the ambiguity introduced by natural language documentation. Stakeholders may interpret descriptions inconsistently, particularly when dealing with nuanced topics like sustainability behaviour or food waste tracking (Ratnam, 2023). Furthermore, most techniques rely on fixed or oversimplified personas that fail to reflect the real-world diversity in household demographics, digital proficiency, and food consumption patterns. These limitations restrict the ability to model needs of both high engagement users (e.g., sustainability advocates) and disengaged users (e.g., elderly donors with low tech literacy).

Another critical shortcoming is the inability of traditional techniques to model **dynamic or predictive behaviours** based on historical usage data. For instance, the system being proposed introduces **analytics powered expiry monitoring** and **gamified environmental feedback**, capabilities that are forward looking and context sensitive. These functionalities require understanding not just what users say they need, but how they interact over time, which conventional elicitation cannot surface on its own (Wang et al., 2024). To overcome these challenges, alternative techniques are proposed.

Table 2: Alternative Techniques

Alternative Technique	Description	Limitation
Data-Driven Persona Enrichment	Enhancing personas using behavioural analytics and user segmentation	Requires large datasets
Sentiment and Review Analysis	Analysing app reviews and forums to extract pain points	Noisy data and low specificity
Real-time Requirement Feedback Loops	Using app analytics to gather in-app usage patterns	Post-deployment focus

These approaches allow analysts to uncover latent needs, such as:

- I. Providing sustainability impact metrics (e.g., carbon savings dashboards),
- II. Enabling expiry-based recipe planning, and
- III. Integrating with local NGOs for real time donation scheduling.

Such features distinguish Urban FoodShare from existing surplus food by supplementing traditional methods with behavioural and analytics driven insights, the system can be tailored to drive long term engagement, sustainability, and community impact, objectives that are otherwise missed or underexplored using standard techniques alone.

4. Proposed Techniques

To capture and model the specific requirements of the Urban FoodShare platform, a risk informed behavioural modelling approach was adopted. This procedure goes beyond conventional functional analysis by systematically identifying, classifying, and mitigating risks inherent in intelligent, socially embedded systems. The approach incorporates insights from literature on AI enabled expiry tracking (Tiwari et al., 2024), user engagement gamification (Santos et al., 2025), and real time stakeholder coordination (Wu & Lin, 2023).

a. What are the risks faced by these systems?

Urban FoodShare, as a data intensive, behaviour sensitive application, faces three primary types of risks:

- **Technical Risks:** Inaccurate expiry predictions due to OCR failures or inconsistent label formats.
- **Operational Risks:** Pickup scheduling breakdowns arising from asynchronous NGO availability.
- **User Risks:** Low retention and motivation stemming from insufficient incentives or complexity in interaction.

b. How do you analyse and classify these risks?

Risks were analysed through domain exploration and system benchmarking. Each risk was categorized using a threefold classification matrix:

- Technical (systemic faults like poor OCR),
- Operational (process and scheduling gaps),
- User centric (motivational or behavioural disengagement).

This classification was informed by prior failures in food sharing systems and gaps identified during literature review and app comparisons (Camilleri, 2025; OLIO, 2023).

c. How do you identify the potential root causes of each identified risk?

Root causes were extracted through:

- Requirement mapping across similar platforms (e.g., OLIO, Flashfood),
- Feature gap analysis,
- Behavioural assumptions from user personas.

For example:

Technical risk stems from non-standard expiry formats.

Operational risk stems from manual NGO logistics.

User risk arises from lack of reward feedback loops and overwhelming workflows for time poor users.

d. How will you eliminate and reduce those risks?

Each risk was mapped to a targeted mitigation strategy:

Table 3: Risk Classification Matrix

<i>Risk Type</i>	<i>Description</i>	<i>Mitigation Strategy</i>
<i>Technical</i>	Inconsistent expiry labels	AI + OCR with fallback to manual entry
<i>Operational</i>	Manual NGO schedule coordination	Real-time API-based NGO calendar integration
<i>User</i>	Lack of motivation, unclear value	Gamification (badges, CO ₂ tracker, social leaderboards)

e. Translate additional requirement specifications from your analysis that will mitigate the risks or defend the system

From this risk analysis, the following system level requirement patterns were derived:

- The system must automate expiry prediction through OCR and allow manual correction for edge cases.
- The system must support real time NGO calendar integration via APIs for seamless donation pickup scheduling.
- The system must include gamification mechanics—usage dashboards, milestone badges, and CO₂ tracking to sustain engagement.

These functional and quality requirements directly mitigate the identified risks and reflect the system's innovative position compared to traditional food sharing apps.

5. Requirements Specification

5.1 Requirements and Systems

This section outlines the core functional, quality, and business requirements for the proposed Urban FoodShare system.

5.1.1 Functional Requirements (FR)

Requirement 1

The system must allow users to input expiry dates via OCR or manual entry.

Rationale: This requirement supports the core functionality of tracking surplus food. OCR ensures automation for barcode scanning, while manual entry serves as a backup, addressing diverse packaging types and varying user tech literacy.

Requirement 2

The system must suggest meal plans based on food nearing expiry.

Rationale: This feature proactively helps reduce waste by encouraging consumption of items before they expire. It also enhances usability by simplifying daily meal planning through contextual recommendations.

Requirement 3

The system must allow scheduling of pickups with local NGOs.

Rationale: Donation scheduling reduces friction in redistribution logistics. This requirement ensures that surplus food reaches NGOs on time, reducing spoilage and supporting community food security.

Requirement 4

The system must maintain user profiles with donation history.

Rationale: Tracking user donations builds accountability and trust with NGO partners while also motivating users through a record of their social and environmental impact.

Requirement 5

The system must display nearby donation points.

Rationale: Offering real time alternatives to pickups ensures no donation opportunity is missed due to unavailability. It enhances flexibility and supports last minute drop offs.

5.1.2 Quality Requirements (QR)

Requirement 6

The system must process expiry inputs within 5 seconds.

Rationale: Fast input processing improves usability and ensures smooth user experience. Delays in scanning or logging may lead to user frustration and decreased engagement.

Requirement 7

The system must encrypt personal and donation data.

Rationale: Protecting sensitive data is essential to maintain user trust and ensure legal compliance with privacy standards like GDPR.

Requirement 8

The system must function offline for basic entry.

Rationale: Users may not always have reliable internet. This requirement allows uninterrupted logging of surplus items, with automatic sync once reconnected, improving accessibility during outages.

5.1.3 Business Requirements (BR)

Requirement 9

The system must integrate with at least 3 NGO platforms.

Rationale: Partnering with multiple NGOs ensures broader coverage and scalability, enabling more households to contribute and receive services efficiently.

Requirement 10

The system must log CO₂ savings per donation.

Rationale: Displaying environmental impact data promotes eco conscious decision making and supports the app's alignment with sustainability goals (e.g., SDG 12.3).

Requirement 11

The system must support reporting for government incentives.

Rationale: In regions offering tax benefits or credits for food donations, users need formal reports. This feature incentivizes regular contributions and strengthens stakeholder engagement.

5.2 Identifying User Requirements

5.2.1 Stakeholders and Primary Users

- **Household User:** Individuals or families aiming to reduce personal food waste.
- **NGO Coordinator:** Personnel from food banks or nonprofits managing surplus food pickups.
- **Council Administrator:** Local government representatives promoting sustainability.
- **System Backend Services:** External systems providing OCR, location, and scheduling services.

5.2.2 Personas

Persona 1: Priya – Sustainability Oriented Professional

Age: 29

Location: Melbourne, VIC

Goals: Reduce personal food waste, donate surplus efficiently

Challenges: Busy schedule, unaware of nearby food donation points

Persona 2: George – Retired Volunteer

Age: 68

Location: Sydney, NSW

Goals: Facilitate food collection for his local food bank

Challenges: Tech hesitance, prefers simple scheduling tools

5.2.3 User Stories

- As a household user, I want to scan expiry dates on my groceries, so that I can track what food needs to be used soon.
- As a logistic coordinator from an NGO, I want to receive automatic notifications of scheduled pickups, so that I can better coordinate NGO deliveries.
- As a Council Administrator, I want reports on food donations, so that I can evaluate community engagement.

5.2.4 Acceptance Criteria

For Priya's user story:

- The system must accept image input from mobile camera.
- OCR must return expiry date in DD/MM/YYYY format.
- User can manually correct scanned items – keeping or not.

5.2.5 Usage Scenario

Actor: *Priya (Household User)*

Environment: Mobile App at Home

Goal: Reduce household food waste by planning meals

Tasks:

1. Priya opens Urban FoodShare app.
2. She scans expiry labels using her phone.
3. App shows items expiring soon.
4. Priya chooses suggested meals.
5. After that she sees she has unused items that will expire within few days and schedules a donation pickup.

5.3 Modelling Requirements

5.3.1 Use Case Diagram

The following use case diagram provides a concise visual representation of how key actors interact with the Urban FoodShare platform.



Figure 3: Use Case Diagram

5.3.2 Detailed Use Case Description

Table 4: Detailed Use Case Description

Use Case Name	Schedule Donation Pickup	
Scenario	Mobile App Interaction - Urban FoodShare App	
Triggering Event	Household user chooses to schedule a food donation pickup before item expiry	
Brief Description	User logs into the Urban FoodShare app, views expiry list, selects food items for donation, and schedules a pickup with a nearby NGO based on availability. The system confirms the schedule, updates both user and NGO calendars, and sends confirmation notifications.	
Primary Actor	Household User	
Related Use Cases	Includes: Login to account, View expiry list Extends: Confirm donation	
Stakeholders	NGO/Food Bank (for pickup coordination), Council Administrator (for reporting)	
Precondition	- User must be logged into the app - Expiry-tracked items must be listed in the system - NGO pickup availability must be accessible	
Postcondition	Pickup successfully scheduled with NGO - Confirmation sent to both user and NGO - Scheduled donation logged into user donation history	
Flow of Activities	Actor	System
	1. User logs into app	1.1 Validate credentials
	2. User navigates to Expiry List	2.1 Display items nearing expiry
	3. User selects items for donation	3.1 Mark items as selected
	4. User clicks "Schedule Pickup"	4.1 Retrieve NGO availability
	5. User selects preferred time slot	5.1 Validate and reserve slot
	6. User confirms donation	6.1 Send confirmation to NGO and user
Exception Conditions	7. User receives pickup details	7.1 Update calendar and history logs
	2a. If user is not logged in: - Prompt login screen.	
	4a. If NGO calendar is not accessible: - Notify user of technical issue and suggest retrying later.	
	5a. If selected time slot is unavailable: - Display alternative slots or allow manual entry.	
	6a. If confirmation fails: - Allow user to retry or save draft to confirm later.	

5.3.3 Activity Diagram

The activity diagram below illustrates the key operational flow of the Urban FoodShare platform from a household user's perspective. It outlines the process from login or registration, through pantry scanning and expiry tracking, to donation scheduling and report generation. The diagram highlights key decision points and actions, offering a clear visual representation of how users interact with the system to reduce household food waste.

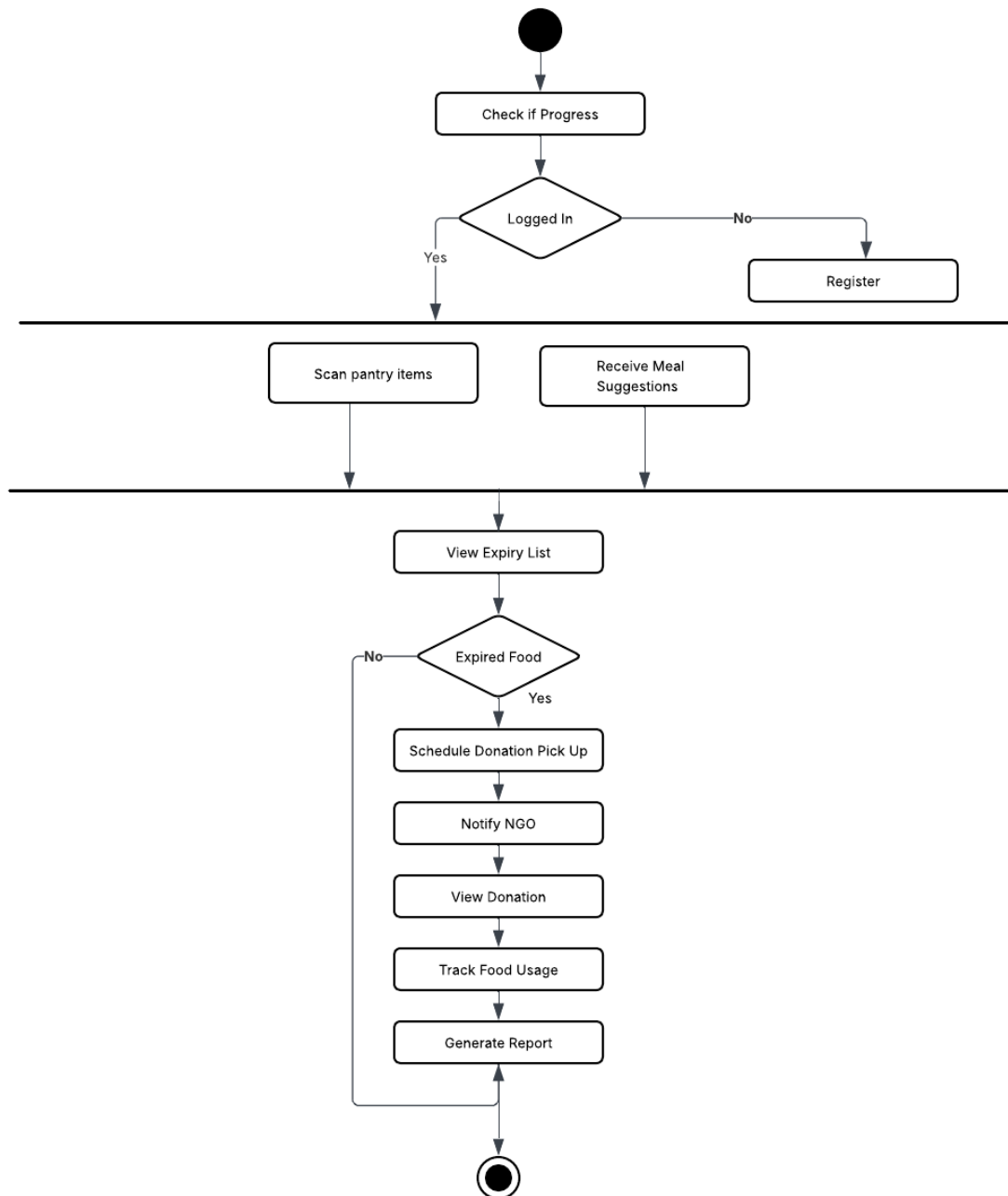


Figure 4: Activity Diagram

5.3.4 Wireframe Diagram

The Urban FoodShare wireframe shows the app interface for households to log, manage, and donate surplus food. Users can scan expiry dates and find donation points, while NGOs and councils use it to coordinate pickups and monitor redistribution.

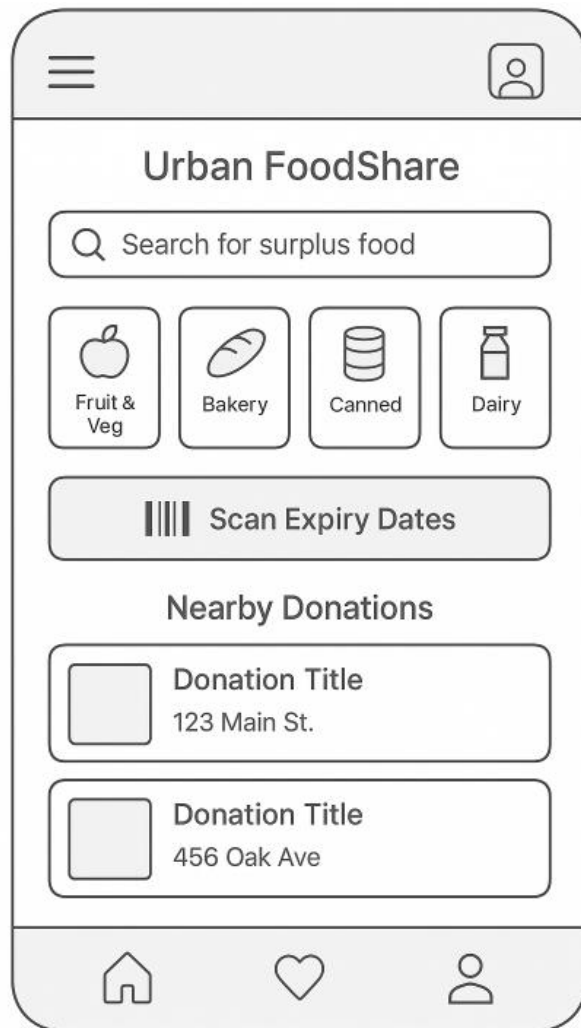


Figure 5: Wireframe Diagram

6. Input and Output Requirement

The table outlines key data elements in Urban FoodShare, including types, examples, and uses. Fields like expiry date and pickup time aid scheduling, while Boolean values support decision logic and ensure consistent input for tracking and reporting.

Table 5: Data Types

Data Name	Data Type	Example	Explanation
Username	Text	Priya Sharma	Sequence of characters
Contact Number	Number/Integer	0400123456	10-digit Australian number
Email Address	Text	priya@email.com	Must be a valid email format
Food Item Name	Text	Bread	Free-text input
Expiry Date	Date	2024-07-15	Stored in YYYY-MM-DD format
Donation Status	Boolean	True	Yes/No or True/False value
Pickup Time	Time	10:00 AM	Standardized time format
NGO Availability	Boolean	False	Yes/No input from NGO dashboard
Recipe Tags	Text/List	Vegan, 10-minute	Multiple entries separated by commas

The table below outlines key validation checks in the Urban FoodShare platform, including existence, data type, and logic-based validations. These checks ensure accurate input, valid expiry dates, and smooth pickup coordination with NGOs.

Table 6: Input Verification

Data Item	Validation Check Type	Details / Justification
<i>User Name</i>	Existence Check	Field must not be empty
<i>Contact Number</i>	Data Type + Range Check	Must be numeric, exactly 10 digits
<i>Email Address</i>	Format + Existence Check	Must include "@" and "." symbols, not empty
<i>Food Item Name</i>	Existence Check	Must not be empty
<i>Expiry Date</i>	Date Format + Future Date Check	Valid date in future
<i>Donation Status</i>	Boolean Check	Must be True/False value
<i>Pickup Time</i>	Time Format Check	Must match HH:MM AM/PM
<i>Carbon Savings</i>	Derived Value Check	System calculated from saved meals
<i>NGO Availability</i>	Boolean Check	Validated through toggle or dropdown
<i>Recipe Tags</i>	Optional Field Check	If present, must be comma-separated strings

7. Conclusion

Urban FoodShare presents a technically feasible, socially valuable, and analytically intelligent approach to reducing household food waste. By leveraging OCR, behavioural analytics, and integration with NGO systems, it addresses logistical, awareness, and user engagement challenges that current apps overlook.

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